

Dataset for Question 7: synthetic_ETA.csv

ABOUT DATASET

synthetic_ETA.csv is a synthetic dataset containing 1,500 observations, created for educational purposes. The variables represent plausible ride-hailing journey characteristics but do not correspond to actual YEGO customer, driver, or journey records. The *observation_id* variable uniquely identifies each record and should not be used as a predictor. The target variable, *eta_minutes*, represents the estimated journey ETA in minutes.

Data description

Variable	Description	Type / Unit
observation_id	A unique identifier assigned to each observation in the dataset. It is used only to identify individual records and should not be used as a predictor.	Integer
journey_distance_km	Distance of the ride from origin to destination.	Numerical, km
traffic_index	A synthetic traffic-congestion score representing the level of traffic during the journey. Higher values indicate heavier traffic.	Numerical, 5–100
time_of_day_hours	Time at which the journey begins, represented as hours on a 24-hour clock. For example, 8.5 represents 08:30.	Numerical, hours
weather_score	Numerical representation of weather conditions affecting the journey. A value close to 0 represents favourable conditions, while a value close to 1 represents more adverse conditions.	Numerical, 0–1
road_slope_pct	Approximate average road gradient along the journey. Positive values indicate uphill slope and negative values indicate downhill slope.	Numerical, %

road_type	Main type of road used during the journey. Categories are local, arterial, express, and highway.	Categorical
previous_journey_time_min	Duration of a previous comparable journey, providing historical information about travel time under similar conditions.	Numerical, minutes
traffic_proxy	A synthetic variable strongly related to the traffic index and included to create a realistic multicollinearity problem.	Numerical
distance_proxy	A synthetic variable strongly related to journey distance and included to create a multicollinearity problem.	Numerical, km-like scale
average_speed_kmh	Estimated average travelling speed during the journey, calculated from prevailing traffic and road conditions. Lower speeds generally indicate greater congestion.	Numerical, km/h
driver_experience_years	Approximate number of years of driving experience of the assigned driver.	Numerical, years
passenger_rating	Historical rating associated with the passenger, on a scale from 1 to 5.	Numerical, 1–5
random_noise_feature	A randomly generated variable with no meaningful relationship to journey ETA. It is included to test whether regularization can reduce the influence of an uninformative predictor.	Numerical
eta_minutes	Estimated duration of the journey in minutes. This is the variable that the regression models are required to predict.	Numerical, minutes